FINITE POINT SETS IN A SEPARATED LOCALLY QUASI-FINITE SCHEME OVER AN AFFINE LIE IN A COMMON AFFINE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12661-12666 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 12668-12675, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion to the collected quasi-finite-separated-quasi-affine card; supplies its finite-set corollary used by the nearby-open series (L16081/L16112).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma quasi-finite-separated-quasi-affine (the ZMT quasi-affineness theorem); Properties, lemma ample-finite-set-in-affine (finite sets in quasi-affine schemes admit affine neighbourhoods).

Source reference. Original lemma 02LR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12426–12429. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12675–12675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a separated, locally quasi-finite morphism with Y affine. Then every finite set of points of X is contained in an open affine of X.*

Proof. Let $x_1, \ldots, x_n \in X$. Choose a quasi-compact open $U \subset X$ with $x_i \in U$. Then $U \to Y$ is quasi-affine by Lemma 02LR. Hence there exists an affine open $V \subset U$ containing $x_1, \ldots, x_n$ by Properties, Lemma 01ZY. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.